Encoding Categorical: One Hot Encoding

Hazem Mohamed
AI Engineer

Encoding Categorical

1. DEFINITION

Hazem Mohamed
AI Engineer

Encoding Categorical

- Encoding Categorical: is used when we have categorical (non numerical data), such as countries: Germany, France, Span
- Note:

Machine learning models cannot work directly with text, sp we convert these categories into numerical form

One Hot Encoding

EXPLAIN WITH EXAMPLE

Hazem Mohamed
AI Engineer

One Hot Encoding

- One Hot Encoding:
  - Instead of assigning numbers like :
    - Germany = 1
    - France = 2
    - Spain = 3
- ❌ This is wrong because it creates a false order
  - (Spain > France > Germany)
- ✅ Correct Approach: One Hot Encoding
  - We create a separate column for each category:

Hazem Mohamed
AI Engineer

One Hot Encoding

Country	Germany	France	Spain
Germany	1	0	0
France	0	1	0
Spain	0	0	1

- Each row has:
  - 1 → The country exists
  - 0 → The country does not exist
- These columns are called dummy variables.
- Why are they called "Dummy Variables?
  - In statistics and machine learning, the term "dummy" doesn't mean "stupid." Instead, it refers to something that is a stand-in or a placeholder for something else.

Hazem Mohamed
AI Engineer

One Hot Encoding

- Final clean Definition:
  - One Hot Encoding is a technique used to convert categorical variable into multiple binary (0 / 1) columns, where each category is represented by a separate columns, allowing machine learning model to process non numeric data without introducing false relationship

Follow for More

AI | Machine Learning | Python

- Simple explanations + Practical examples

Follow for more

Hazem Mohamed
AI Engineer